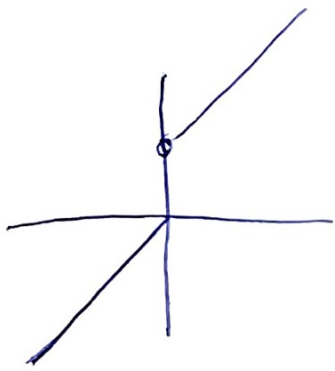
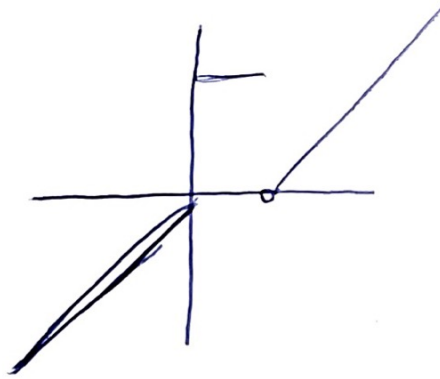




$$f(x) = \begin{cases} x & x \leq 0 \\ x+1 & x > 0 \end{cases}$$

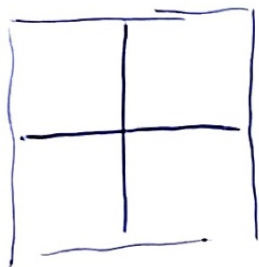


$$g(x) = \begin{cases} x & x \leq 0 \\ x-1 & x > 1 \\ 3 & 0 < x \leq 1 \end{cases}$$

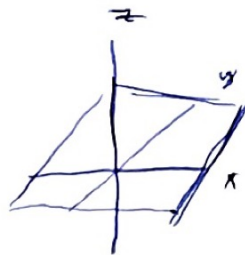
injective,
but not linear, continuous
or surjective

We have $g(f(x)) = x$, so g is a left inverse for f .

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^3$$



f



$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

f has a left inverse: (b/c injective)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$g \quad f$

identity on $\mathbb{R}^2$ ✓

but no right inverse: (b/c not surjective)

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} a & c & e \\ b & d & f \\ 0 & 0 & 0 \end{pmatrix}$$

$f \quad h?$

not the identity on $\mathbb{R}^3$, regardless of entries of h .

X